

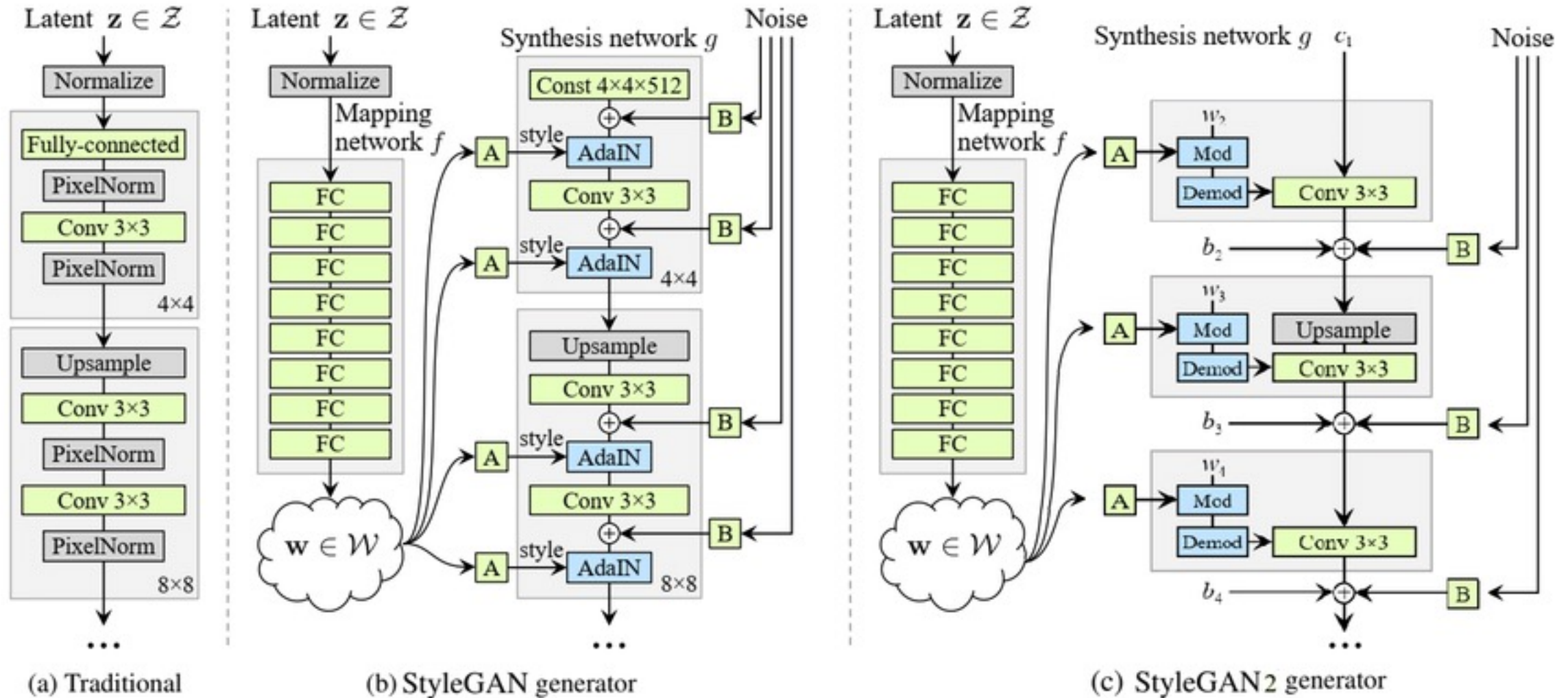
StyleSpace Analysis

(CVPR 2021 oral)

김상현, 고흥권, 허다운, 조경진, 전선영, 김준철

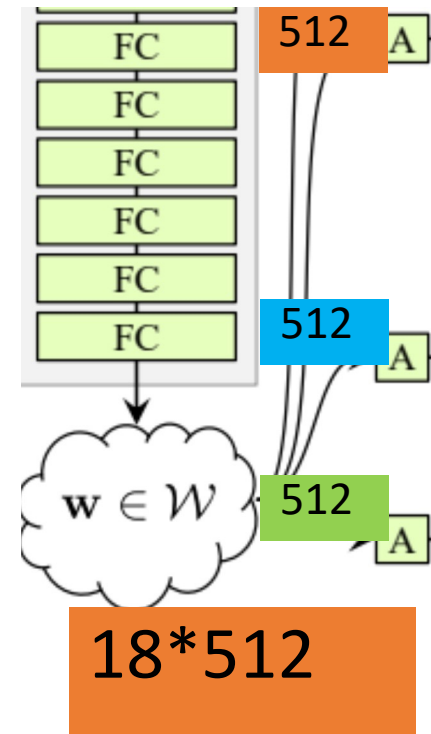
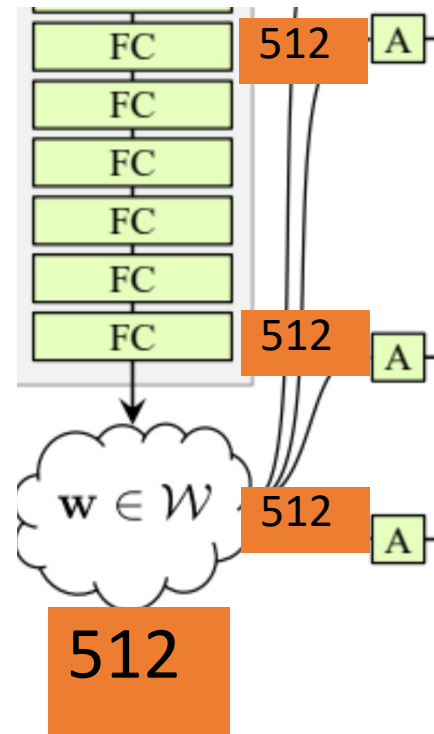
발표: 고흥권

Background: StyleGAN, StyleGAN2

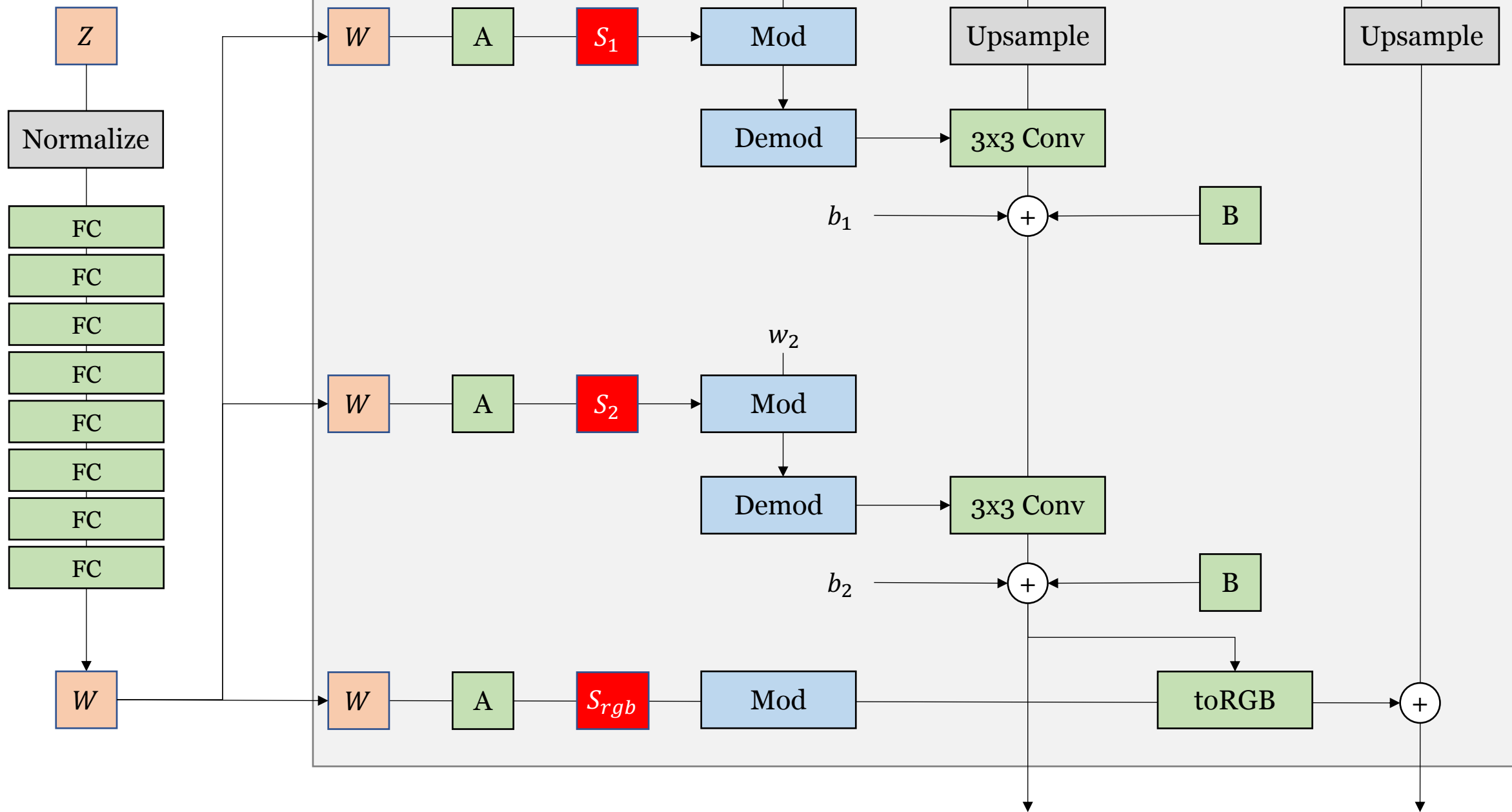


Background: Latent Manipulation

- W space $\rightarrow$ 512 dim
- $W+$ space $\rightarrow 18 * 512 = 9216$ dim



Key idea: StyleSpace (S space)



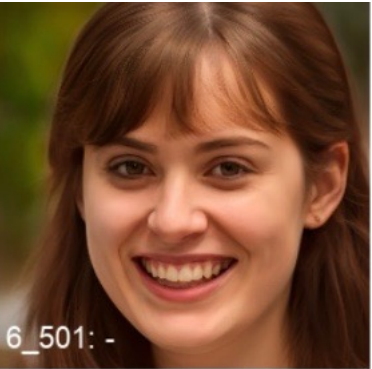
W space $\rightarrow$ 512 dim
W+ space $\rightarrow$ 9216 dim
S space $\rightarrow$ 9088 dim

$\mathcal{W}+$ layer index	$\mathcal{S}$ layer index	resolution	layer name	type	# channels
0	0	4×4	Conv	s_1	512
1	1	4×4	ToRGB	s_{tRGB}	512
2	2	8×8	Conv0_up	s_1	512
3	3	8×8	Conv1	s_2	512
3	4	8×8	ToRGB	s_{tRGB}	512
4	5	16×16	Conv0_up	s_1	512
5	6	16×16	Conv1	s_2	512
5	7	16×16	ToRGB	s_{tRGB}	512
6	8	32×32	Conv0_up	s_1	512
7	9	32×32	Conv1	s_2	512
7	10	32×32	ToRGB	s_{tRGB}	512
8	11	64×64	Conv0_up	s_1	512
9	12	64×64	Conv1	s_2	512
9	13	64×64	ToRGB	s_{tRGB}	512
10	14	128×128	Conv0_up	s_1	512
11	15	128×128	Conv1	s_2	256
11	16	128×128	ToRGB	s_{tRGB}	256
12	17	256×256	Conv0_up	s_1	256
13	18	256×256	Conv1	s_2	128
13	19	256×256	ToRGB	s_{tRGB}	128
14	20	512×512	Conv0_up	s_1	128
15	21	512×512	Conv1	s_2	64
15	22	512×512	ToRGB	s_{tRGB}	64
16	23	1024×1024	Conv0_up	s_1	64
17	24	1024×1024	Conv1	s_2	32
17	25	1024×1024	ToRGB	s_{tRGB}	32

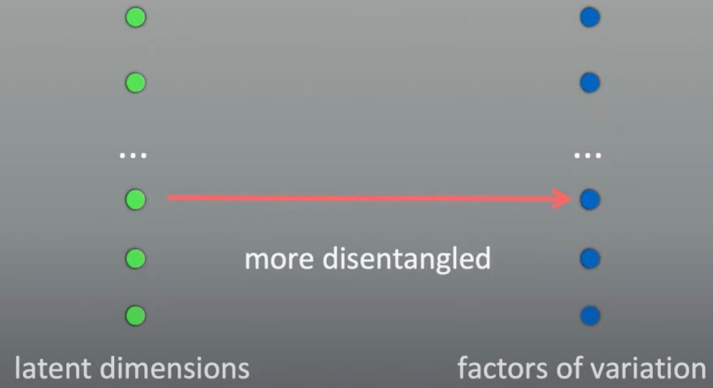
Table 2. Breakdown of StyleSpace channels by generator layers.

Contributions: Disentangled controls

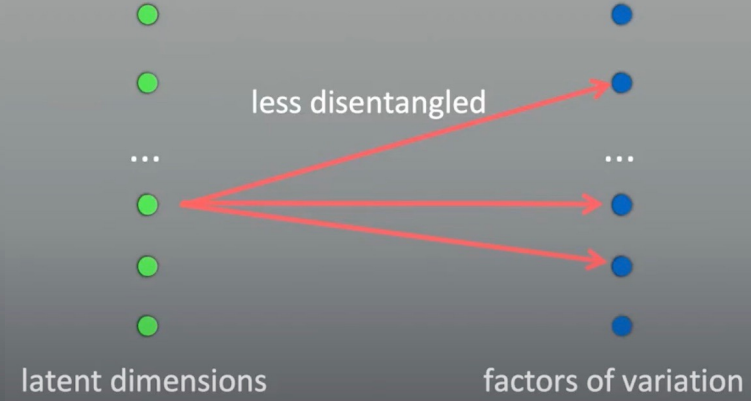
- Evaluation using DCI metrics
- D (Disentanglement)
- C (Completeness)
- I (Informativeness)



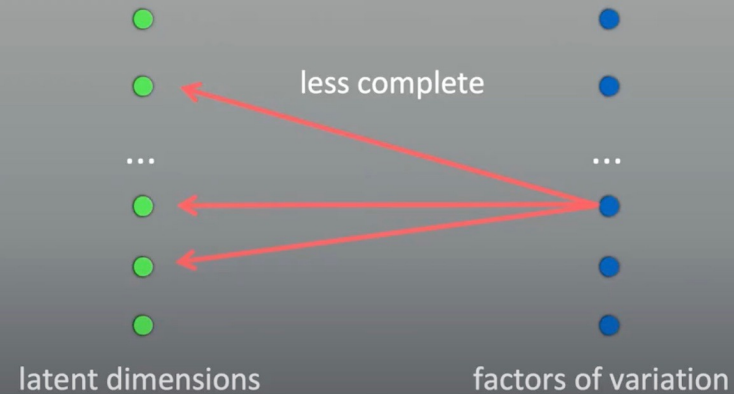
Disentanglement



Disentanglement



Completeness

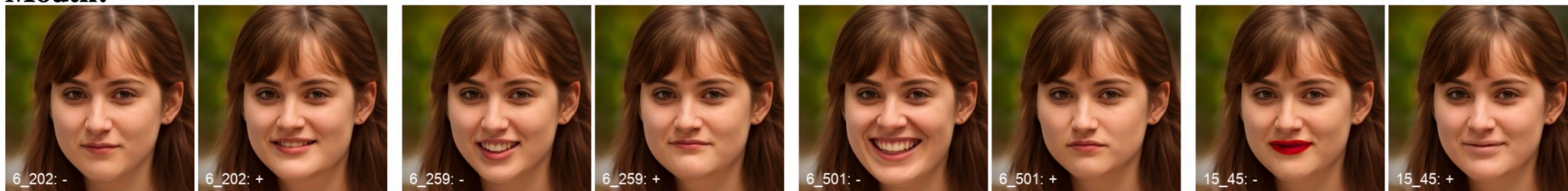


Completeness

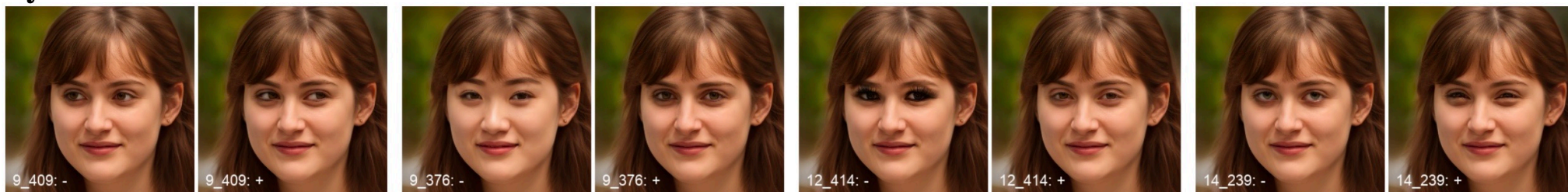


Sample Results

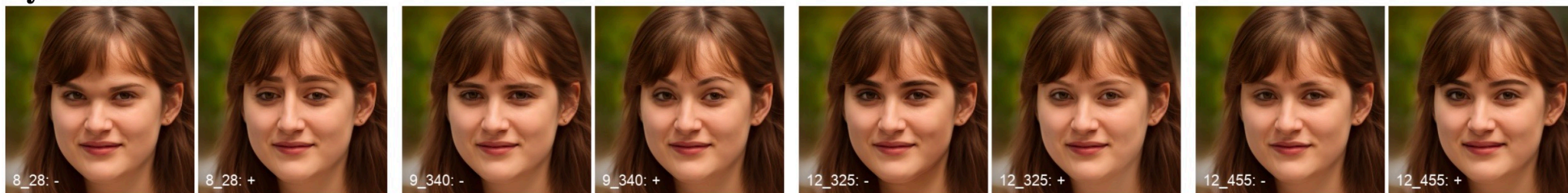
Mouth:



Eyes:



Eyebrows:

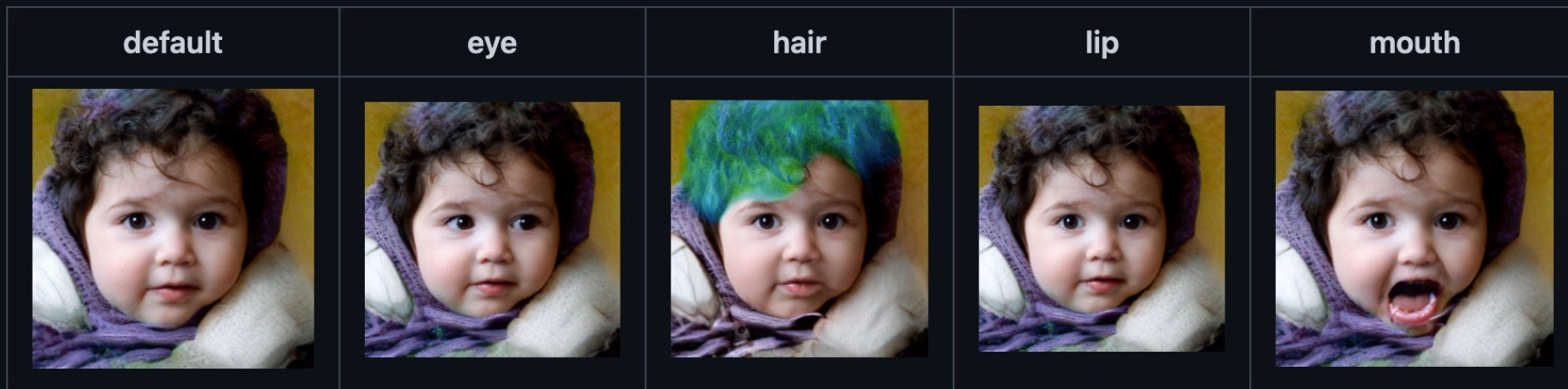


Ears:



Simple StyleSpace Run

 [Open in Colab](#)



Run StyleSpace

1. locate `stylegan2-ffhq-config-f.pt` under the folder `checkpoint`
2. run `python stylespace.py`

Credits

StyleGAN2 model and implementation:

<https://github.com/rosinality/stylegan2-pytorch>

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License (MIT) <https://github.com/rosinality/stylegan2-pytorch/blob/master/LICENSE>

StyleSpace pytorch implementation

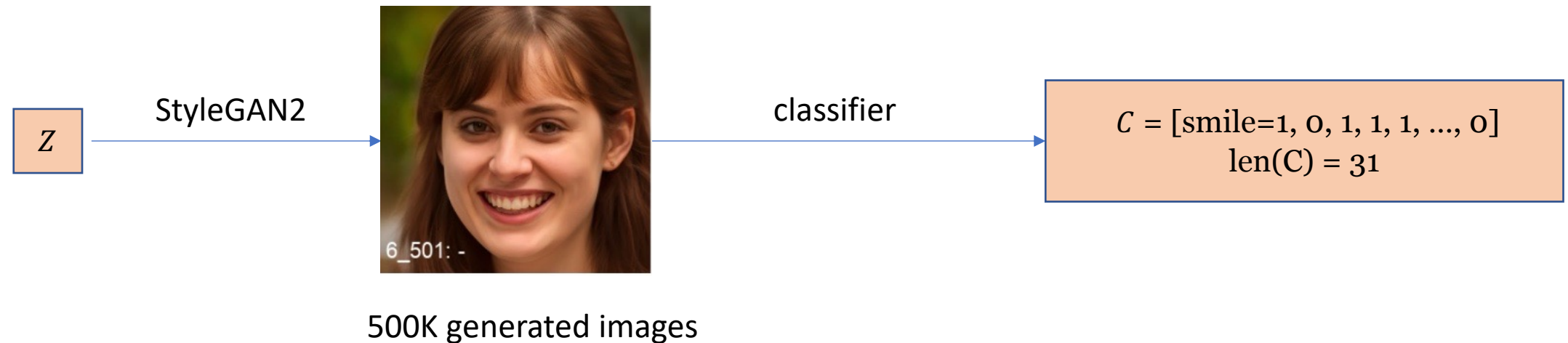
<https://github.com/xrenaa/StyleSpace-pytorch>

<https://github.com/hyungkwonko/StyleSpace-pytorch>

- Any questions?

3. Disentanglement of StyleGAN latent spaces

- StyleGAN2 trained on FFHQ
- 40 binary classifiers pretrained on CelebA attributes
- Used only 31 attributes



3. Disentanglement of StyleGAN latent spaces

(Using random sampling)

	Comparison w/ $\mathcal{Z}$ and $\mathcal{W}$		
	Disent.	Compl.	Inform.
$\mathcal{Z}$	0.31	0.21	0.72
$\mathcal{W}$	0.54	0.57	0.97
$\mathcal{S}$	0.75	0.87	0.99

(Using inversion)

	Comparison with $\mathcal{W}+$		
	Disent.	Compl.	Inform.
$\mathcal{W}+$	0.54	0.64	0.94
$\mathcal{S}$	0.63	0.81	0.98

Table 1. Disentanglement, completeness and informativeness for different latent spaces (larger is better, maximum is 1). The two comparisons are performed using different sets of images; thus, the scores are not comparable between the two tables.

4. Detecting locally-active style channels

- Find locally-active style channels using **gradient map**
- Backprop to compute gradient map
- Overlap btwn. **semantic region** and **gradient map**

4. Detecting locally-active style channels

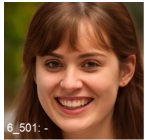
$$OC_{u,c}^s = \frac{|(G_u^s > t_{u,c}^s) \cap M_c^s|}{|M_c^s|^d}.$$

How much overlap???



5. Detecting attribute-specific channels

- the differences between the mean style vectors
 - the positive examples (exemplar mean)
 - the entire generated distribution (population mean)



$$\delta^e = \frac{s^e - \mu^p}{\sigma^p}$$

Normalized style vector

δ^e
 δ^e
 δ^e
 δ^e
 δ^e

μ^e and σ^e

$$\theta_u = \frac{|\mu_u^e|}{\sigma_u^e}$$

u = style channel
9088 dim

5. Detecting attribute-specific channels

- Advantage: Requires only 10-30 positive examples
- Advantage2: do not require negative samples

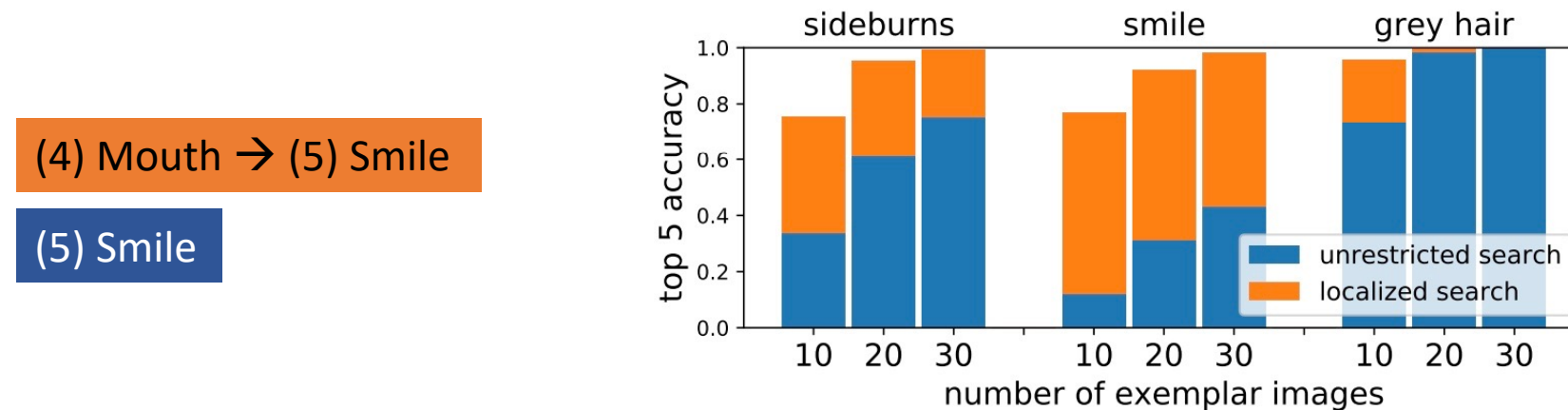
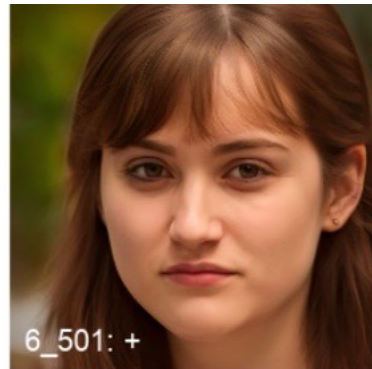


Figure 5. Top-5 detection accuracy for attribute-specific controls (for three target attributes) using 10, 20, or 30 positive examples.

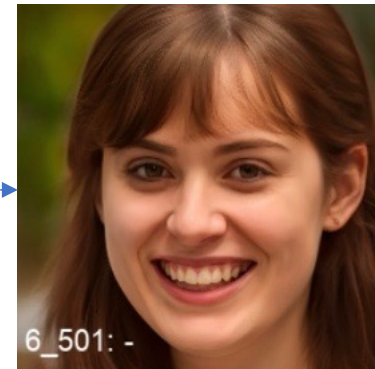
6. Disentangled attribute manipulation

- Comparison to previous works
- Attribute Dependency (AD)
 - Quantify independent manipulation
 - Change of logits



$C=[\text{smile}=0, 1, \dots]$

$\text{Logit}=[\text{smile}=0.02, 0.78, \dots]$



$C=[\text{smile}=1, 1, \dots]$

$\text{Logit}=[\text{smile}=0.98, 0.83, \dots]$

6. Disentangled attribute manipulation

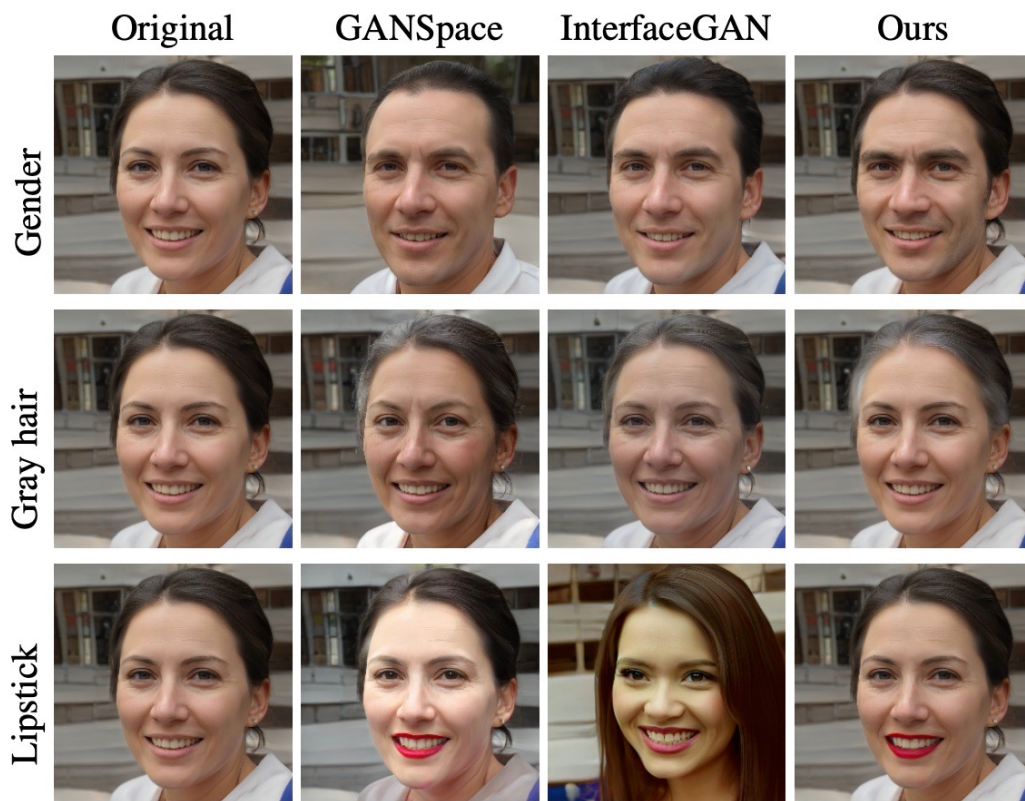


Figure 6. Comparison with state-of-the-art methods using the same amount of manipulation $\Delta l_t = 1.5\sigma(l_t)$.

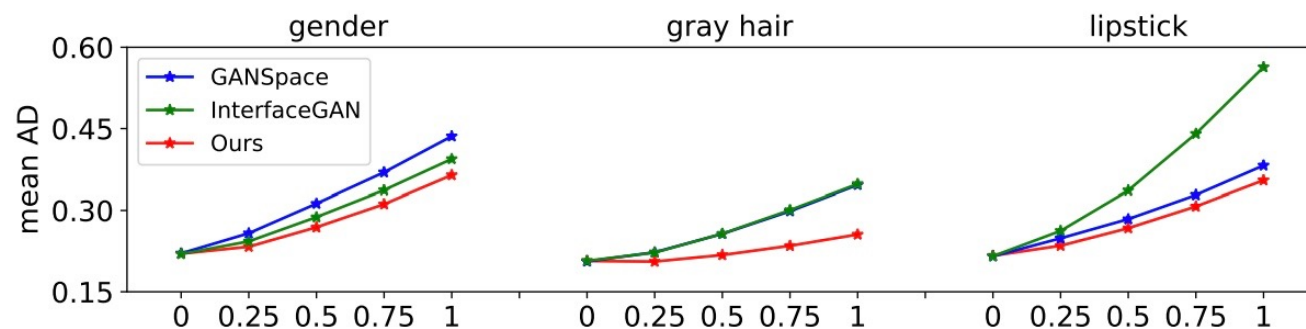


Figure 8. Mean-AD vs. the degree of target attribute manipulation ($\Delta l_t / \sigma(l_t)$). Lower mean-AD indicates better disentanglement.

7. Real-image editing (using In-domain GAN)

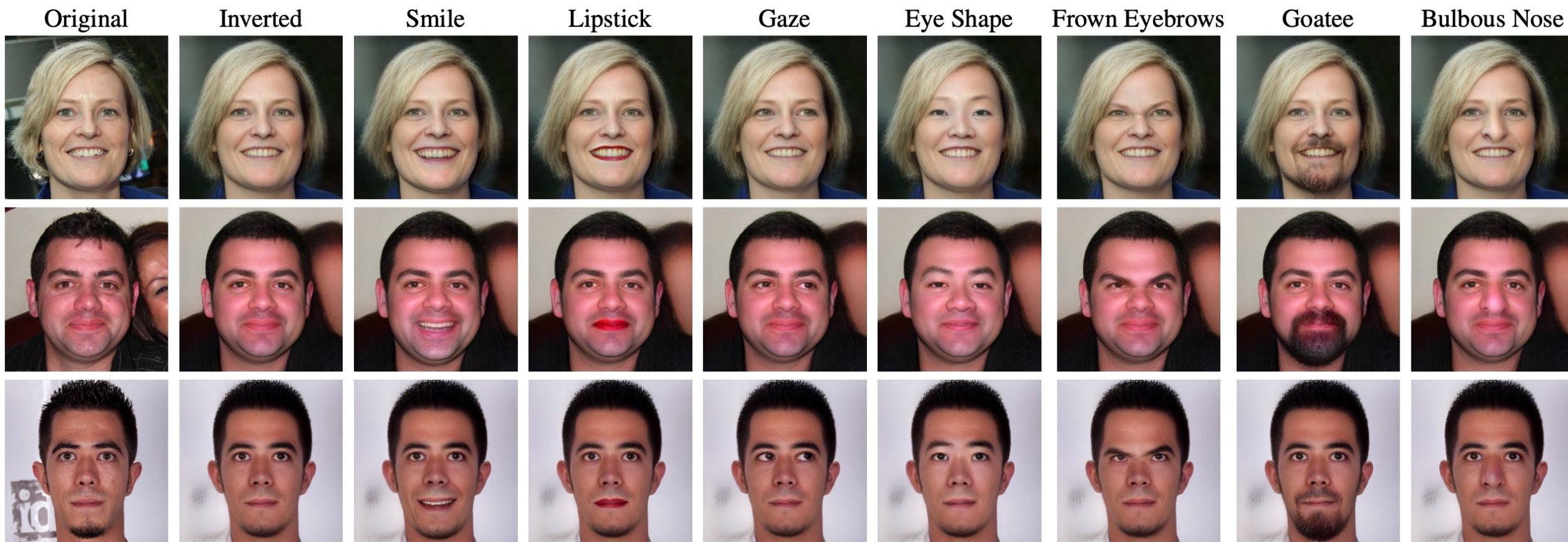
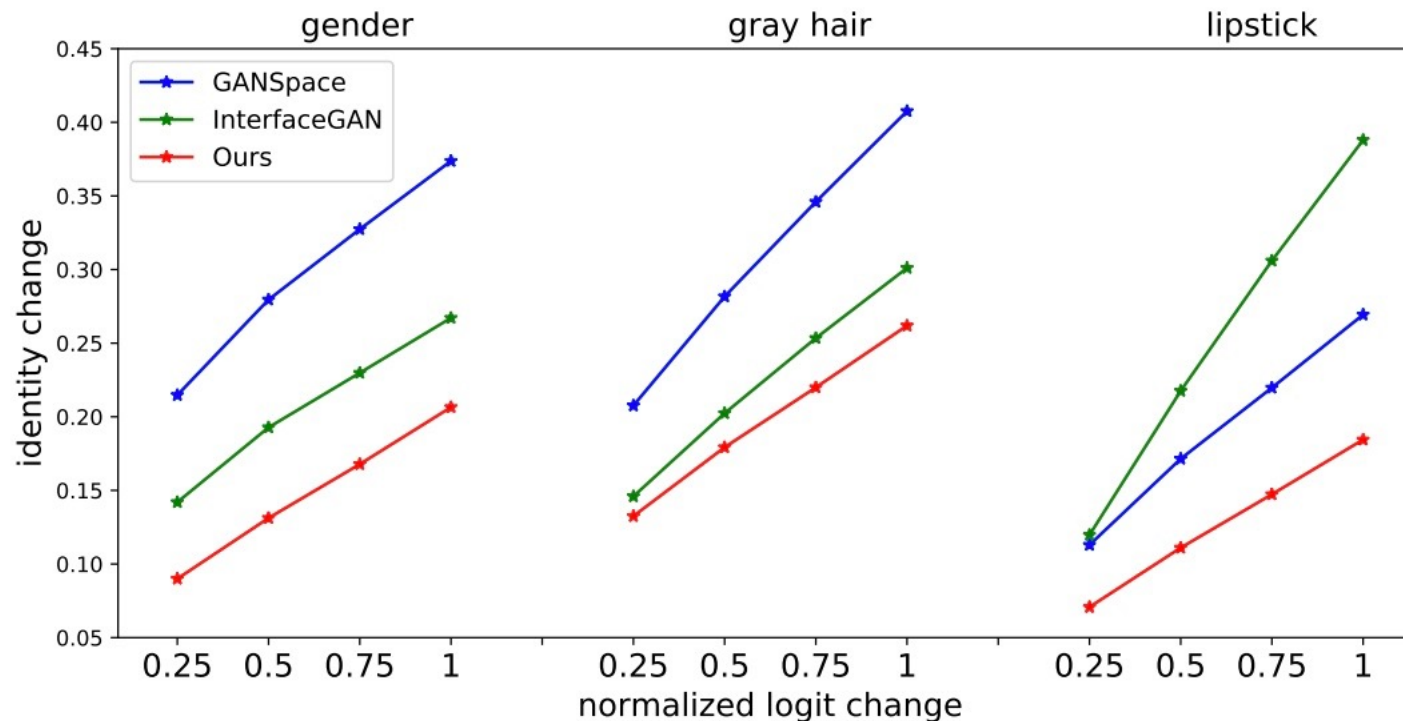


Figure 7. Manipulation of real images using encoder-based inversion. Original images are from FFHQ, and were not part of the encoder's training set. More results can be found in app. Figure 20.

Less ID change (supp)

- ID changes less than baselines (e.g., GANSpace, InterfaceGAN)
- Euclidean distance between Features when embedded using FaceNet



- Any questions?